

Real GDP and Inflation

Session 3 – Introduction to Macroeconomics

Francesco Chiocchio

ESCP Business School

Recap: Session 2

10 minutes. Answer these questions silently, on your own; then we go through the answers.

GDP and national accounts

1. What is GDP and what are the three ways of measuring it?
2. What are the four components of GDP? Give a brief description of each.
3. What is the difference between a stock and a flow? Give an example.
4. How do we move from gross domestic product (GDP) to net national income (NNI)?

The closed-economy, long-run model

5. **Supply:** What are the factors of production? How are they combined?
6. **Demand:** What determines consumption, investment, and government purchases? What happens to net exports?
7. Which variable adjusts to equilibrate supply and demand?
8. In which markets do we have equilibrium?
9. What happens if taxes decrease, *ceteris paribus*?

Recap: Answers – GDP

1. GDP is the market value of all final goods and services produced within an economy during a given period. Three ways to measure it:
 - ▶ **Production:** sum value added.
 - ▶ **Expenditure:** $Y = C + I + G + NX$.
 - ▶ **Income:** sum the incomes generated by production.
2. The four expenditure components are:
 - ▶ consumption: household spending on newly produced goods and services, excluding new housing;
 - ▶ investment: new capital, new housing, and inventories; buying stocks or bonds is not investment in GDP;
 - ▶ government purchases: goods and services, not transfers;
 - ▶ net exports: exports minus imports.
3. A **stock** is measured at a point in time (wealth); a **flow** is measured over a period (income per year).

Recap: From GDP to National Income

Question: How do we move from GDP to GNI and NNI?

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Domestic versus national

- ▶ GDP counts production **within the country's borders**.
- ▶ GNI counts income earned by the country's **residents**.

$$\text{GNI} = \text{GDP} + \text{net income from abroad}$$

Gross versus net

- ▶ Gross income includes the capital used up in production.
- ▶ Net income subtracts this depreciation.

$$\text{NNI} = \text{GNI} - \text{depreciation}$$

Source: UK Office for National Statistics, *Blue Book 2016*, 2015 data; figure from supplied teaching materials.

UK GDP by category of expenditure, GNI and NNI: 2015

	£million
Consumption expenditure of households and NPISH (C)	1 216 113
Government final consumption (G)	360 828
Gross capital formation (I)	327 855
Exports of goods and services (X)	510 340
less Imports of goods and services (M)	-548 908
Statistical discrepancy	3 332
Gross domestic product (GDP) (at market prices)	1 869 560
<i>plus</i> Net income from abroad	-37 016
Gross national income (GNI)	1 832 544
<i>less</i> Capital consumption (depreciation)	245 144
Net national income (NNI)	1 587 400

Recap: The Closed-Economy Model

Supply

$$Y^s = F(\bar{K}, \bar{L}) = \bar{Y}$$

- ▶ The factors of production are capital K and labor L ; the production function F represents technology.
- ▶ In the benchmark model, K , L , and F are fixed.
- ▶ Output is therefore fixed at $\bar{Y}$.

Demand

$$Y^d = C(\bar{Y} - \bar{T}) + I(r) + \bar{G}$$

- ▶ Consumption depends on disposable income.
- ▶ Investment falls when the real interest rate rises.
- ▶ G and T are exogenous policy variables.

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In equilibrium, demand equals the fixed supply of output:

$$\bar{Y} = C(\bar{Y} - \bar{T}) + I(r) + \bar{G}.$$

The adjusting variable is r : the real interest rate moves until desired spending equals fixed output.

Recap: Goods and Financial Markets

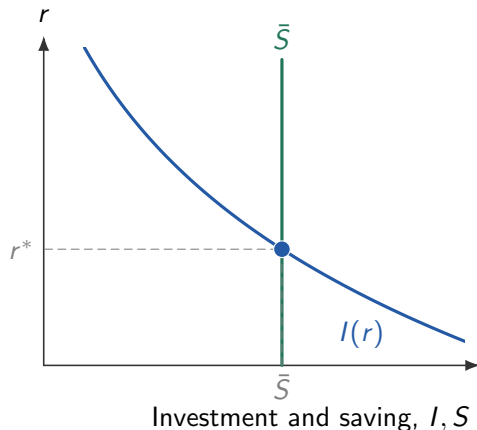
Rearrange goods-market equilibrium:

$$\underbrace{\bar{Y} - C(\bar{Y} - \bar{T}) - \bar{G}}_{\bar{S}} = I(r).$$

$$S_{\text{private}} = \bar{Y} - \bar{T} - C(\bar{Y} - \bar{T}),$$

$$S_{\text{public}} = \bar{T} - \bar{G},$$

$$S = S_{\text{private}} + S_{\text{public}}.$$



Goods-market equilibrium and loanable-funds equilibrium are the same condition: $\bar{S} = I(r)$.

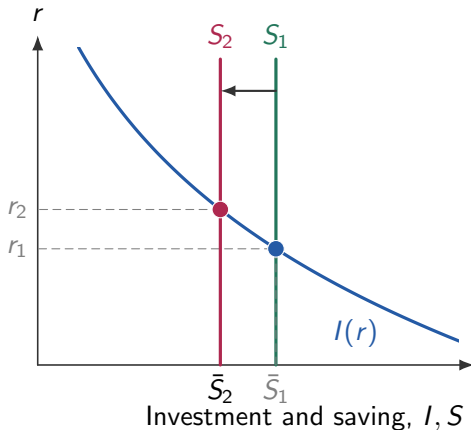
Recap: What Does a Tax Cut Do?

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Recap: What Does a Tax Cut Do?

Question: Suppose $\bar{T}$ falls while $\bar{G}$ and $\bar{Y}$ remain fixed. What happens?

- ▶ Disposable income rises, so consumption rises.
- ▶ Private saving rises by only the part of the tax cut that households do not consume.
- ▶ Public saving falls by the full tax cut. National saving therefore falls.
- ▶ The supply of loanable funds shifts left: r rises and investment falls.



Session 3: Outline

1. **Real Versus Nominal GDP**

Separate changes in quantities from changes in prices.

2. **Measuring Inflation**

Construct the GDP deflator, CPI, and PCE deflator.

3. **Comparing Measures and Special Cases**

Scope, weights, core inflation, and real values.

4. **Costs and Benefits of Inflation**

Expected inflation, unexpected inflation, and moderate inflation.

Each measure combines available data with assumptions about what to include and how to weight it.

Reading: Mankiw, *Macroeconomics*, 12th ed., Chapters 2 and 6.

1. Real Versus Nominal GDP

Prices, quantities, and the choice of a base year

One Change, Two Possible Explanations

Suppose nominal GDP rises by 10 percent. Has the economy produced 10 percent more?

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Not necessarily. Nominal GDP can rise because:

1. the economy produces more goods and services;
2. prices rise; or
3. both quantities and prices rise.

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To measure production, we need to hold prices constant.

Nominal GDP and Real GDP

For goods $i = 1, \dots, n$, let $P_{i,t}$ be the price and $Q_{i,t}$ the quantity produced in year t .

Nominal GDP

$$NGDP_t = \sum_{i=1}^n P_{i,t} Q_{i,t}$$

Values current production at current prices.

Real GDP, base year B

$$RGDP_t^{(B)} = \sum_{i=1}^n P_{i,B} Q_{i,t}$$

Values current production at base-year prices.

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Values current production at base-year prices.

Only real GDP isolates changes in the quantity of output.

Try It: Nominal and Real GDP

Use 2019 as the base year.

1. Compute nominal GDP in each year.
2. Compute real GDP using 2019 prices.
3. Which series better measures output growth?

	2019		2020		2021	
	<i>P</i>	<i>Q</i>	<i>P</i>	<i>Q</i>	<i>P</i>	<i>Q</i>
Good A	\$30	900	\$31	1,000	\$36	1,050
Good B	\$100	192	\$102	200	\$100	205

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Good A	\$30	900	\$31	1,000	\$36	1,050
Good B	\$100	192	\$102	200	\$100	205

Year	Nominal GDP	Real GDP at 2019 prices
2019	\$46,200	\$46,200
2020	51,400	50,000
2021	58,300	52,000

Nominal GDP grows by 26.2 percent; real GDP grows by 12.6 percent. **Real GDP gives the clearer picture of output growth.**

Real GDP Controls for Price Changes

Between 2019 and 2021:

$$\frac{58,300 - 46,200}{46,200} \times 100 = 26.2\% \quad (\text{nominal GDP growth})$$

$$\frac{52,000 - 46,200}{46,200} \times 100 = 12.6\% \quad (\text{real GDP growth})$$

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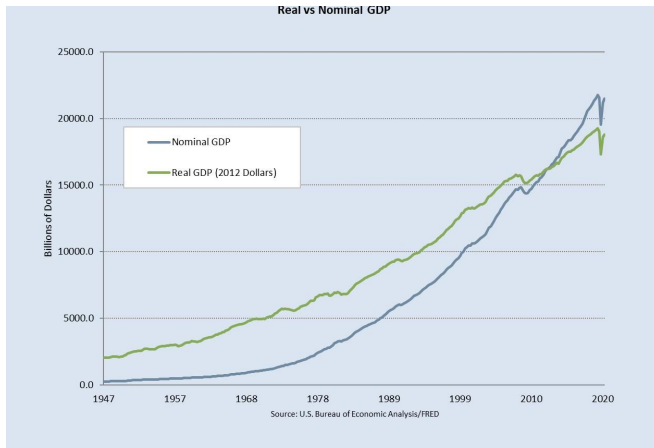
$$\frac{58,300 - 46,200}{46,200} \times 100 = 26.2\% \quad (\text{nominal GDP growth})$$

$$\frac{52,000 - 46,200}{46,200} \times 100 = 12.6\% \quad (\text{real GDP growth})$$

- ▶ Nominal growth combines changes in quantities and prices.
- ▶ Real growth holds prices fixed, so it reflects changes in production.

The difference between nominal and real growth contains information about prices.

Nominal and Real GDP in the Data



Nominal and real GDP coincide in the base year. Away from that year, their gap reflects changes in the price level.

Source: U.S. Bureau of Economic Analysis, via Federal Reserve Economic Data (FRED); chart from the supplied Session 3 slides, 1947–2020.

Base Years Versus Chain Weights

Problem: Fixing a base year can give an unrepresentative picture of real GDP growth when relative prices change.

Example: computer prices have fallen sharply while college tuition has risen. Valuing today's computers and education at prices from 20 or 30 years ago would give them obsolete weights.

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Chain-weighted real GDP

1. Measure growth between two adjacent years using prices from those years.
2. Combine the two nearby-year growth measures so neither year alone determines the weights.
3. Link these growth rates across years to form a real GDP series—hence “chain weighted.”

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Why use it? The base year changes continuously, so the weights keep up with changes in relative prices and the composition of production. The U.S. Bureau of Economic Analysis therefore emphasizes chain-weighted real GDP.

Trade-off: chain weighting is more representative, but harder to calculate than fixed-base-year real GDP.

2. Measuring Inflation

GDP deflator, consumer price index, and PCE deflator

Price Level Versus Inflation

- ▶ A **price index** summarizes the overall level of prices relative to a base period.
- ▶ The **inflation rate** is the percentage change in that index.

For any price index P_t ,

$$\pi_t = 100 \times \frac{P_t - P_{t-1}}{P_{t-1}}.$$

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$$\pi_t = 100 \times \frac{P_t - P_{t-1}}{P_{t-1}}.$$

- ▶ An index of 120 does not mean inflation is 120 percent.
- ▶ It means the measured price level is 20 percent above the base-period level, which is normalized to 100.
- ▶ Inflation depends on how the index changes from one period to the next.

There Is No Single Price

An economy has millions of prices. Some rise quickly, some slowly, and some fall.

	Price change	Share of spending or output
Rent	+5%	large
Coffee	+20%	small
Laptop computers	-10%	?

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To measure the **overall** price level, we need a weighted average:

- ▶ Which prices belong in the average?
- ▶ What weight should each price receive?
- ▶ Should the weights remain fixed or change over time?

Different answers produce the GDP deflator, CPI, and PCE deflator.

Measure 1: The GDP Deflator

The GDP deflator compares the current-price value of domestic production with the value of the same production at base-year prices:

$$\text{GDP deflator}_t = 100 \times \frac{NGDP_t}{RGDP_t^{(B)}}.$$

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The GDP deflator compares the current-price value of domestic production with the value of the same production at base-year prices:

$$\text{GDP deflator}_t = 100 \times \frac{NGDP_t}{RGDP_t^{(B)}}.$$

It covers all final goods and services produced domestically:

- ▶ consumer goods and services;
- ▶ investment goods;
- ▶ government services;
- ▶ exports, but not imports.

Try It: GDP Deflator and Inflation

Complete the final two columns.

Year	Nominal GDP	Real GDP	GDP deflator	Inflation
2019	\$46,200	\$46,200		n.a.
2020	51,400	50,000		
2021	58,300	52,000		

1. Compute the GDP deflator in each year.
2. Compute inflation from 2019 to 2020 and from 2020 to 2021.

GDP Deflator and Inflation: Answers

Year	Nominal GDP	Real GDP	GDP deflator	Inflation
2019	\$46,200	\$46,200	100.0	n.a.
2020	51,400	50,000	102.8	2.8%
2021	58,300	52,000	112.1	9.1%

GDP Deflator and Inflation: Answers

Year	Nominal GDP	Real GDP	GDP deflator	Inflation
2019	\$46,200	\$46,200	100.0	n.a.
2020	51,400	50,000	102.8	2.8%
2021	58,300	52,000	112.1	9.1%

For 2021,

$$\pi_{2021} = 100 \times \frac{112.1 - 102.8}{102.8} \approx 9.1\%.$$

The deflator is a price index; its percentage change is an inflation rate.

Inside the GDP Deflator: Two-Good Example

With two goods and base year B , let

$$RGDP_t = Q_{1,t}P_{1,B} + Q_{2,t}P_{2,B}.$$

$$\begin{aligned}\frac{\text{GDP deflator}_t}{100} &= \frac{NGDP_t}{RGDP_t} = \frac{Q_{1,t}P_{1,t} + Q_{2,t}P_{2,t}}{RGDP_t} \\ &= \frac{Q_{1,t}P_{1,t}}{RGDP_t} + \frac{Q_{2,t}P_{2,t}}{RGDP_t} \\ &= \frac{P_{1,t}}{P_{1,B}} \underbrace{\frac{Q_{1,t}P_{1,B}}{RGDP_t}}_{\text{weight of good 1}} + \frac{P_{2,t}}{P_{2,B}} \underbrace{\frac{Q_{2,t}P_{2,B}}{RGDP_t}}_{\text{weight of good 2}}.\end{aligned}$$

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- ▶ The deflator is a weighted average of price changes relative to the base year.
- ▶ The weights are base-price shares of **current production**, so they change when current quantities change.

A Useful General Rule

For moderate percentage changes,

$$\% \Delta(XY) \approx \% \Delta X + \% \Delta Y, \quad \% \Delta \left(\frac{X}{Y} \right) \approx \% \Delta X - \% \Delta Y.$$

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$$\% \Delta (XY) \approx \% \Delta X + \% \Delta Y, \quad \% \Delta \left(\frac{X}{Y} \right) \approx \% \Delta X - \% \Delta Y.$$

Because

$$\text{GDP deflator} = 100 \times \frac{NGDP}{RGDP},$$

if nominal GDP grows by 9 percent and real GDP grows by 4 percent, inflation is approximately 5 percent.

Measure 2: The Consumer Price Index

The consumer price index (CPI) measures the cost of a basket purchased by a representative household.

1. Survey households to determine the basket and its expenditure weights.
2. Collect prices for the goods and services in that basket.
3. Compare the current cost of the basket with its base-period cost.

$$CPI_t = 100 \times \frac{\text{cost of the basket at time } t}{\text{cost of the basket in base period } B}.$$

Measure 2: The Consumer Price Index

The consumer price index (CPI) measures the cost of a basket purchased by a representative household.

1. Survey households to determine the basket and its expenditure weights.
2. Collect prices for the goods and services in that basket.
3. Compare the current cost of the basket with its base-period cost.

$$CPI_t = 100 \times \frac{\text{cost of the basket at time } t}{\text{cost of the basket in base period } B}.$$

The CPI tracks household living costs, not the price of all domestic production.

Try It: Compute the CPI

The fixed basket contains 20 pizzas and 10 records. Use 2018 as the base year.

Year	Pizza price	Record price	Cost	CPI	Inflation
2018	\$10	\$15			
2019	11	15			
2020	12	16			
2021	13	15			

For each year, compute the cost of the basket, the CPI, and inflation since the previous year.

Try It: Compute the CPI

The fixed basket contains 20 pizzas and 10 records. Use 2018 as the base year.

Year	Pizza price	Record price	Cost	CPI	Inflation
2018	\$10	\$15	\$350	100.0	n.a.
2019	11	15	370	105.7	5.7%
2020	12	16	400	114.3	8.1%
2021	13	15	410	117.1	2.5%

For example,

$$CPI_{2020} = 100 \times \frac{20(12) + 10(16)}{20(10) + 10(15)} = 114.3.$$

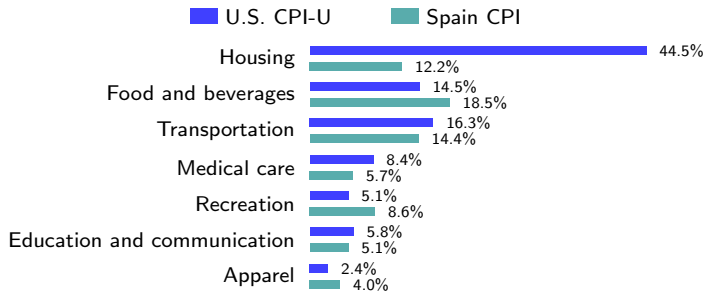
The basket costs 14.3 percent more than in 2018.

CPI Weights Differ Across Countries (2025)

Which categories receive the largest weights?

- ▶ Rank housing, food, transportation, medical care, recreation, and apparel.
- ▶ Would the ranking be the same in Spain and the United States?
- ▶ Would it remain the same twenty years from now?

CPI Weights Differ Across Countries (2025)



Sources: BLS, CPI-U, Dec. 2025; INE, Spain CPI, 2025. Selected basket weights, not a harmonized breakdown. U.S. housing includes owners' equivalent rent (26.2%) and furnishings; Spain excludes imputed rent and groups furnishings separately. U.S. food includes restaurant meals and alcohol; the Spanish bar covers food and non-alcoholic beverages. Spain's education and communication groups are combined here.

Inside the CPI: Two-Good Example

Let $C_{i,B}$ be the quantity of good i in the base-period basket, and let $Basket_t$ be the current cost of that fixed basket. Then

$$\begin{aligned}\frac{CPI_t}{100} &= \frac{Basket_t}{Basket_B} = \frac{P_{1,t}C_{1,B} + P_{2,t}C_{2,B}}{P_{1,B}C_{1,B} + P_{2,B}C_{2,B}} \\ &= \frac{P_{1,t}}{P_{1,B}} \underbrace{\frac{P_{1,B}C_{1,B}}{Basket_B}}_{\text{weight of good 1}} + \frac{P_{2,t}}{P_{2,B}} \underbrace{\frac{P_{2,B}C_{2,B}}{Basket_B}}_{\text{weight of good 2}}.\end{aligned}$$

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- ▶ The CPI is also a weighted average of price changes.
- ▶ Its weights reflect base-period consumer expenditure shares.
- ▶ Between basket updates, these weights **remain fixed**.

CPI and GDP Deflator: Three Differences

1. **Scope:** the deflator covers all domestic output; the CPI covers household purchases. A rise in the price of machinery bought by firms affects the deflator, not the CPI.
2. **Imports:** an imported car affects the CPI but not the domestic GDP deflator.
3. **Weights in our fixed-base examples:** CPI uses a fixed basket (Laspeyres); the GDP deflator uses current quantities (Paasche). Official series also use updated baskets and chain weighting.

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Orange-crop example: a frost destroys most of the crop, so orange output collapses and the price of the few remaining oranges jumps.

CPI: oranges retain their base-period weight. The index can overstate the cost-of-living increase because consumers substitute apples for oranges.

GDP deflator: the current orange weight becomes tiny. The index can understate consumers' loss from expensive oranges.

Why a Fixed-Basket CPI Can Overstate Inflation

1. **Substitution:** consumers shift toward goods whose relative prices fall, but a fixed basket cannot fully reflect that response.
2. **New goods:** new products expand choice before they receive an appropriate weight in the basket.
3. **Quality change:** part of a higher price may pay for a better product rather than the same product at a higher price.

Why a Fixed-Basket CPI Can Overstate Inflation

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Each problem can make measured CPI inflation exceed the rise in a true cost-of-living index. Statistical agencies update baskets and adjust for quality to reduce these biases, but their exact size remains uncertain.

Measure 3: The PCE Deflator

The personal consumption expenditures (PCE) deflator measures prices for household consumption:

$$\text{PCE deflator}_t = 100 \times \frac{\text{nominal consumer spending}_t}{\text{real consumer spending}_t}.$$

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- ▶ Like the CPI, it focuses on consumption and includes imported consumer goods.
- ▶ Like the GDP deflator, its weights adjust as spending patterns change.
- ▶ It therefore captures substitution more quickly than a fixed-basket index.

3. Comparing Measures and Special Cases

Scope, weights, core inflation, and real values

Every Inflation Measure Makes Choices

To construct an index, statisticians must decide:

1. **Scope:** which goods, services, and people does the index cover?
2. **Weights:** how important is each price, and how often do weights change?
3. **Quality:** how much of a price change reflects a better or worse product?
4. **Data:** which prices and quantities can be observed, and when?

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These measures target the same broad concept—changes in the overall price level—but answer slightly different questions.

GDP Deflator, CPI, and PCE Deflator

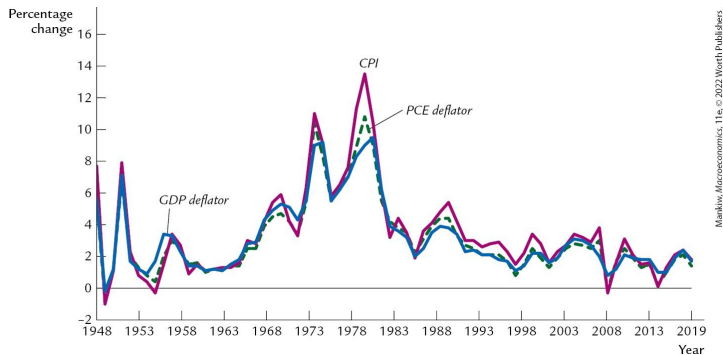
	GDP deflator	CPI	PCE deflator
Question	Prices of domestic output?	Cost of a household basket?	Prices of household consumption?
Scope	All final goods and services produced domestically	Goods and services purchased directly by households	Household consumption, including some spending made on households' behalf
Imports	Excluded	Included	Included
Investment and government output	Included	Excluded	Excluded
Weights	Change with domestic production	Relatively fixed between basket updates	Change with consumption patterns
Main inputs	National accounts data on domestic output	Household expenditure survey for weights; sampled retail prices	National accounts, business, and administrative spending data

GDP Deflator, CPI, and PCE Deflator

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Investment and government output	Included	Excluded	Excluded
Weights	Change with domestic production	Relatively fixed between basket updates	Change with consumption patterns
Main inputs	National accounts data on domestic output	Household expenditure survey for weights; sampled retail prices	National accounts, business, and administrative spending data

There is no universally “best” index. The right measure depends on the question. All three combine many data sources, but the CPI requires a separate survey to construct the household basket.

Why the Measures Usually Move Together



Mankiw, *Macroeconomics*, 11e, © 2022 Worth Publishers

The measures share many prices, so their broad movements are similar. They differ when coverage or weights matter—for example, when import prices or consumption patterns change sharply.

Source: Mankiw, *Macroeconomics*, 11th ed. (2022), Worth Publishers; 1948–2019.

The Measurement Lesson

Inflation is one concept, but it is not one directly observed number.

- ▶ The GDP deflator asks about prices of domestic production.
- ▶ The CPI asks about the cost of a household basket.
- ▶ The PCE deflator asks about prices of household consumption with changing weights.
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When two inflation numbers differ, first ask what each index includes, how it weights prices, and which data it uses.

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- ▶ **Example: fuel and diesel.** Diesel is an input into transport and production, so a fuel-price shock can pass through to food, construction, and many other prices.

What does it miss? Households still buy food and energy. Core inflation is a diagnostic measure, not a replacement for the cost of living.

Both CPI and PCE inflation can be reported in headline and core versions.

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- **Remember this distinction:** nominal and real interest rates return when we study money and monetary policy.

Comparing Money Values Across Time

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Example: If the CPI rises from 108 to 505.4, then \$200 in the first year has the same measured purchasing power as

$$200 \times \frac{505.4}{108} \approx \$936$$

in the later year.

The result inherits the CPI's basket and measurement assumptions.

4. Costs and Benefits of Inflation

Expected inflation, unexpected inflation, and why moderate inflation can help

A Common Misconception: Inflation Reduces Real Wages

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The key question is whether nominal wages adjust with the price level.

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The distinction is whether people had time to adjust.

Expected inflation

- ▶ People can adjust contracts and nominal interest rates.
- ▶ Costs remain because holding money, changing prices, taxes, and accounting are affected.

Unexpected inflation

- ▶ Existing nominal contracts cannot adjust immediately.
- ▶ Purchasing power is redistributed between parties.
- ▶ Uncertainty makes planning harder.

Unexpected inflation is usually more disruptive: it adds redistribution and uncertainty to the costs that remain even when inflation is anticipated.

Costs of Expected Inflation

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5. **Accounting and planning costs:** nominal values from different dates become harder to compare.

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General rule

- ▶ If $\pi > \pi^e$, the realized real interest rate $i - \pi$ is lower than expected. Borrowers gain and lenders lose.
- ▶ If $\pi < \pi^e$, the realized real interest rate is higher than expected. Lenders gain and borrowers lose.

High Inflation Also Raises Uncertainty

When inflation is high and variable, future inflation is harder to predict.

- ▶ Long-term contracts become riskier.
- ▶ Relative-price signals become harder to interpret.
- ▶ Saving and investment decisions become more difficult.
- ▶ Arbitrary redistributions become more frequent and larger.

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The problem is not only that prices rise. People also become less certain about how fast prices will rise.

Benefit 1: A Buffer Against Deflation

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A small positive inflation rate provides a buffer against deflation and gives monetary policy more room to cut real interest rates. We return to this in the sessions on monetary policy.

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Economists describe this as inflation **“greasing the wheels” of the labor market**. This benefit does not imply that higher inflation is always better.

What to Remember

- ▶ **Real GDP isolates output changes.** Nominal GDP combines prices and quantities; inflation is the percentage change in a price index.
- ▶ **GDP deflator, CPI, and PCE cover different prices.** Their scopes and weights reflect different questions.
- ▶ **Every inflation measure depends on data and assumptions.** Check baskets, substitution, new goods, and quality before interpreting a number.
- ▶ **Expected and unexpected inflation have different costs.** Surprises also redistribute purchasing power; moderate inflation can ease wage adjustment and buffer deflation.

Next session: why output fluctuates, and how price adjustment separates the short run from the long run.